

# Understanding Attenuation

From German brewing and more

This article is intended to give the advanced home brewer a better understanding of attenuation, how it is affected by the brewing process and how controlling it can be used to improve the brewing process.

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## Calculating Attenuation

Attenuation refers to the percentage of original extract that has been fermented:

$$\text{Attenuation} = 100 \% * (\text{starting extract} - \text{current extract}) / (\text{starting extract})$$

This formula works with extract given in weight percentages or degree Plato. Extract refers to all the non water substances (sugars, dextrins, proteins, vitamins, minerals, etc.) that are present in the wort. The percent extract or Plato scale is a measure in percent of how much of the wort's weight is comprised of extract. Since, at least in the wort and beer gravities that most brewers work with, an almost linear relationship between (specific gravity - 1) and extract percentages exists, the above formula can also be expressed as:

$$\text{Attenuation} = 100 \% * (\text{starting gravity} - \text{current gravity}) / (\text{starting gravity} - 1)$$

for brewers who prefer to work with specific gravity.

## Brewing Process and Wort Composition

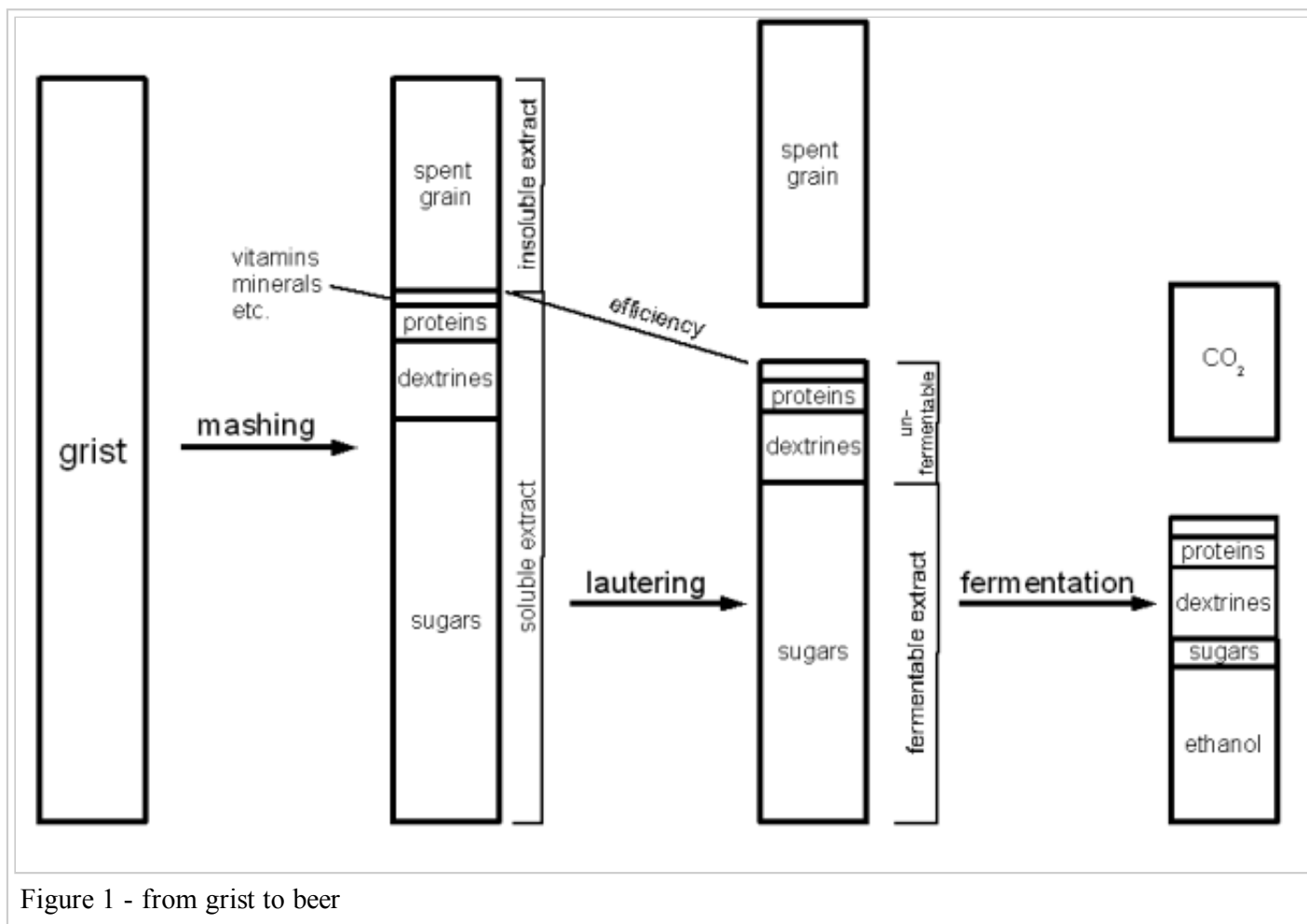


Figure 1 - from grist to beer

To understand the different forms of attenuation we need to take a look at the extract composition first. During mashing, the majority of the grist is converted into water soluble compounds. This is shown in Figure 1.

The conditions during the mashing process as well as the used malts and adjuncts will determine the exact ratio between the various compounds (sugars, dextrines, proteins and others). Once the conversion is complete, the sweet wort is separated from the spent grain in the lauter. Due to inefficiencies during this process not all of the dissolved extract ends up in the boil kettle. The percentage of the total dissolved extract and the extract that actually ends up in the boil kettle is called lauter efficiency and is part of the brewhouse efficiency (See Understanding Efficiency).

During the boil only minor changes happen to the wort composition. The denaturization of the enzymes finally fixes the ratio between fermentable and unfermentable extract. The coagulation of proteins changes the amount and composition of proteins in the wort. Hops will add additional compounds. But these are of little interest for the discussion of attenuation.

During fermentation the fermentable sugars are converted into almost equal parts of CO<sub>2</sub> and ethanol as well as much smaller amounts of other compounds (esters, higher alcohols). The yeast will absorb most of the simple proteins and nutrients. But not all of the fermentable sugars may have been fermented at the time the beer finished and consumed. The amount of fermentable sugars left in the beer has an effect on the beer character and different styles of beer oftentimes have different amounts of fermentable sugars left in them.

## Apparent vs. Real Extract

Hydrometers are calibrated for measuring the extract (sugar, dextrans, proteins and other) content of a water solution through changes in the liquids density in relation to pure water. This calibration is true for wort which only contains water and extract. But when used to measure the extract of beer, which contains ethanol in addition to water, the reading will be skewed by the lower specific gravity of the ethanol. As a result the hydrometer shows a lower density or specific gravity than what would be true if the alcohol would be replaced with water. This measured extract value is called apparent extract (as opposed to the real extract that is measured when there is no alcohol present) and is commonly used when referring to the extract (or specific gravity) of beer. Like the real extract it can be expressed as weight percent, degree Plato or specific gravity. To determine the real extract one can boil-off the alcohol and replace it with distilled water before using a hydrometer. Or, if the original extract is known, the following formula can be used to calculate the real extract from the apparent extract [Realbeer]:

$$\text{real extract} = 0.1808 * \text{original extract} + 0.8192 * \text{apparent extract}$$

## Apparent vs. Real Attenuation

When the apparent extract of the beer is used to calculate its attenuation the result is called apparent attenuation. The use of the real extract will give the real attenuation. When brewers speak of just attenuation they most likely mean apparent attenuation since it can easily be calculated from hydrometer readings. As shown in the Appendix the following relationship exists between real and apparent attenuation:

$$\text{Real Attenuation} = 0.82 * \text{Apparent Attenuation}$$

The real attenuation is only needed when there is an interest in the actual percentage of extract that was fermented or can be fermented. The latter is important for sugar calculations for various priming (carbonation) sugars. Its application is demonstrated in Accurately Calculating Sugar Additions for Carbonation

## Limit of Attenuation

The limit of attenuation is the sum of all the sugars, that the yeast is able to ferment, expressed as a percentage of the total extract content. In other words: it is the attenuation of the beer if no fermentable sugars would be left. There are slight differences in the types of sugars that can be fermented by ale yeast (*S. cerevisiae*) and lager yeast (*S. pastorianus*). In addition to the sugars that can be fermented by ale yeast lager yeast also fully ferments raffinose (ale yeast only ferments this sugar partially) and melibiose [BABC]. Raffinose is absent in brewing wort and melibiose is only present in very small amounts [Alexander 2007]. Conflicting information exists on whether the amount that is present is significant. Given that the differences between the sugars that can be fermented by ale and lager yeast that actually present in the wort are small, the limit of attenuation should have little dependency on the type of yeast that is used.

This fact is taken advantage of in a Fast Ferment Test. This test uses a sample of wort and pitches it with a large amount of yeast. To ensure complete fermentation, in addition to the high pitching rate the sample is kept warm and the yeast is regularly roused (shaking or stirring). Once the fermentation is complete the apparent extract is measured and the limit of attenuation for that wort can be calculated. This is a measure

of the fermentability that was achieved through the brew house processes (mainly mashing and sugar additions in the kettle) and it serves as an upper limit to the attenuation that can be achieved with brewers yeast. It provides the attenuation potential for the fermentation process.

## Yeast Strain Differences in Attenuation

*If there are no limit-of-attenuation differences between the different yeast strains, why are more and less attenuative yeast strains available?*

Though yeast strains are able to ferment all the sugars in the beer, they usually don't get to. In contrast to a fast ferment test, beer is generally fermented at lower temperatures, with smaller pitching rates and without constant rousing. Because of that the yeast may not get a chance to ferment all fermentable sugars in the wort. Flocculation will cause it to slow down its metabolism and drop to the bottom or collect on the surface where it doesn't have as much contact with the wort sugars anymore. Because of nutrient depletion and or high alcohol levels cells may also die before they get a chance to ferment every last bit of sugar in the wort. The result is left over fermentable sugars that play an important role in the character of the finished beer. The closer a beer's attenuation is to its limit the drier and less sweet it will taste. When looking at the attenuation ranges given for commercial yeast you will notice that the less flocculating a given yeast is, the more attenuative it will be. This makes sense since poorly flocculating yeasts will remain in contact with the wort and act as active fermenters for a longer time. The beech wood aging process used by Anheuser-Bush to brew Budweiser does exactly that without relying on poor flocculation alone; it maximizes the contact area between beer and yeast.

[Narziss, 2005] lists ranges for the differences between finished beer attenuation and limit of attenuation for some German beer types:

- Helles : 2 - 4%
- Export : 0.5 - 2 %
- Pilsner : 0.5 - 4 %
- Bock, Dunkel : up to 6 %

Example: A Helles with a target attenuation difference of 3% should be brewed. The wort has an original extract of 12.0 \*P and a fast ferment test final extract of 2.0 \*P. The resulting limit of attenuation is  $100\% * (12 - 2.0) / 12 = 83\%$ . The target attenuation of the finished beer is  $83\% - 3\% = 80\%$ . This attenuation is reached when the beer has an apparent extract of  $12 - 12 * 80\% / 100\% = 2.4$  \*P. But hitting this number exactly is difficult. One way would be to regularly measure the extract of the lagering beer and pull it off the yeast into a serving keg when the desired final extract is reached.

Even if a brewer doesn't go to this extend of control over the beer, performing a fast ferment test and comparing the limit of attenuation to the current attenuation can be helpful in understanding the taste character of a beer. It shows if an unexpected high finishing gravity is due to problems during fermentation (low attenuation but high limit of attenuation) or due to problems during the mashing process (the limit of attenuation is also low and there is little difference to the actual beer attenuation).

## Attenuation Numbers Listed for Yeast Strains

As mentioned above, choice of yeast strain also affects the actual attenuation of the beer. As a guideline yeast vendors like Wyeast (<http://www.wyeastlab.com/default.cfm>) and White Labs (<http://www.whitelabs.com/index-old.html>) list attenuation ranges with their yeasts. But since the yeast strain is only one factor in attenuation (other important factors are mashing and yeast health) these attenuation values are only useful to compare yeasts with each other and cannot be used to predict the final extract (or final gravity) of the beer. This can only be done with a forced ferment test.

I was also told by Wyeast that there is no standard wort for measuring attenuation and that the attenuation levels given are more based on previous performances of that yeast.

In addition to that, one of the yeast banks for commercial brewers (Hefebank Weißenstephan in Germany: Yeast strain characteristics (<http://www.hefebank-weißenstephan.de/page/Downloads/yeast%20strains%20characteristics.pdf>)) doesn't even list attenuation estimates for their yeast. The only indication they are giving is what difference between limit of attenuation and actual beer attenuation can be expected if the fermentation is done properly.

## Affecting Attenuation

When designing a recipe, or using an existing recipe, the brewer generally has a targeted attenuation (original and final extract/gravity) in mind. Unfortunately there is no formula that can be used to calculate this attenuation upfront, though some recipe design software attempts this. Such software generally uses the attenuation given for the selected yeast which may not be the attenuation that you will actually be getting.

As shown above, there are 2 parameters regarding attenuation that the brewer can affect. The first one is the limit of attenuation and the second one is the difference between the final attenuation and the limit of attenuation. The limit of attenuation is set by the wort production and the difference between the final attenuation and the limit of attenuation is set by fermentation. Since the composition of fermentable sugars (mainly the ratio between glucose, maltose and maltotriose) also affects the fermentation performance, mashing also has a small impact on the attenuation difference.

### wort production

- **all grain brewers:**
  - **saccharification rest temperature:** This is the first factor that comes to mind for all grain brewers. For a single step saccharification rest, the mash temperature has a great effect on the fermentability of the resulting wort. The lower the temperature (within a given range of course) the longer the beta-amylase and limit dextrinase will be able to work and produce fermentable sugars. See Starch Conversion. In the Limit of attenuation experiment it was found that, at a saccharification rest temperature above the temperature for maximum fermentability, an increase of the rest temperature by 1 C leads to a limit of attenuation drop of 4%.
  - **mash schedule:** the choice of mash schedule also affects the fermentability. Some beta amylase and limit dextrinase activity is already present during a protein rest and the time it takes to heat to the saccharification rest. In German brewing a 2 stage conversion rest (first stage around 63 C (145F) and the 2nd stage around 70C (160F)) is generally used. In this case the attenuation can also be controlled by the length of the 63C rest which is also called maltose rest. The longer that rest is held the more fermentable sugars can be produced by beta-amylase and limit dextrinase. Another mash schedule factor is the length of the mashing time and the time the wort spends below 175 °F (80 °C). Below that temperature the alpha amylase is still

active and can produce fermentable sugars, though not as effective and quickly as the beta amylase which is quickly denatured at temperatures above 156 °F (70 °C). See Starch Conversion.

- **water to grist ratio:** the enzymatic activity of the amylases is affected by the thickness of the mash. However, during the Limit of attenuation experiment it was observed that mash thickness has no significant impact on the fermentability. This is likely the result of a balance of two factors: thin mashes cause enzymes to be denatured faster but they also cause enzymes to work more efficiently, thus being able to produce the same amount of fermentable sugars even though they aren't active as long.
  - **grain bill composition (base malt):** mashes rich in enzymes, i.e. high diastatic power, (Pilsner malt, Pale malt) will produce more fermentable worts since they contain a larger amount of beta-amylase and limit dextrinase which can produce more maltose than mashes with lower diastatic power (Munich malt or large amounts of unmalted grains) assuming the same saccharification rest temperature.
  - **grain bill composition (specialty malts):** crystal and roasted malts add mostly unfermentable sugars to the wort which lowers its overall fermentability.
  - **mash pH:** the beta and alpha amylase enzymes have different optimal pH ranges (beta amylase : 5.0 - 5.5 pH; alpha amylase : 5.3 - 5.8 [Palmer 2006]) and therefore the mash pH can affect the activity balance between these enzymes. See Starch Conversion. In Limit of attenuation experiment it was found that an attenuation optimum exists between pH 5.3 and 5.7 (measured at room temp). For this and other reasons this pH range should be the mash pH target.
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- **extract brewers:**
    - **type of extract:** different extracts have different levels of fermentability: Get to know the extract that you are using
    - **specialty malts:** specialty malts like crystal and roasted malts add mostly unfermentable sugars thus lowering the fermentability.
    - **blending extracts:** If a lot of control over the fermentability of the extract is desired, the brewer can blend highly fermentable and less fermentable extracts to achieve a desired limit of attenuation. Though this is more predictable than achieving the desired fermentability through mashing most extract brewers rather start all grain brewing before blending extracts
    - **adding unfermentable sugars:** malto dextrin and lactose (milk sugar) are examples of unfermentable sugars that can be added to the wort to make it less fermentable.
  - **partial mash:** this is basically a mix between all grain and extract brewing. The more of the final extract is produced by mashing the more the factors shown for all grain will matter for the fermentability of the wort.

## fermentation

- **lager vs. ale:** lager yeasts are able to completely ferment raffinose and melibiose which ale yeast cannot or only ferment partially. This effect is assumed to be very minimal due to the absence of raffinose and very small amounts of melibiose in brewer's wort. But ale yeasts are slower in their uptake of maltotriose. This results in higher levels of maltotriose left behind by ale yeast.
- **yeast strain:** less flocculant yeasts will remain in suspension longer which gives them the ability to

ferment more of the available sugars. Rousing a flocculant yeast can improve the attenuation if that is desired.

- **yeast health:** healthy yeasts are able to withstand the increasingly toxic environment (ethanol is toxic for yeasts and they rely on healthy cell walls to keep it out of their cells) better than weak ones. This is especially important for high gravity beers where a large amount of healthy yeast is needed to ferment the beer before the high alcohol level starts to affect too many of the yeast cells.
- **pitching rate:** The more yeast is pitched the quicker it can ferment more fermentable sugars in wort. But more is not always better. Overpitching, which can happen when harvested yeast is used, can lead to negative impacts on the flavor of the beer as the reduced yeast growth will change the flavor compounds produced by the yeast. A common rule of thumb for ales is 0.75 million cells per milliliter of wort per degree Plato for ales and 1.5 million cells per milliliter of wort per degree Plato for lagers [Zainasheff].
- **fermentation temperature:** Higher temperatures accelerate the yeast's metabolism and the yeast will be able to consume the sugars faster and generally more complete. But the production of unwanted flavor compounds at higher temperatures limits the fermentation temperature. With good yeast health and sufficient pitching rate the fermentation temperature can be kept fairly low while still ensuring sufficient attenuation. A temperature rise towards the end of fermentation can be beneficial to attenuation while avoiding the off-flavors that higher fermentation temperature early in the fermentation can create.
- **agitation:** If the yeast is roused regularly, more of the yeast cells will be in contact with unfermented sugars and the yeast will be able to ferment the beer faster and more complete. But since this oftentimes goes against the flavor characteristics desired from a yeast strain, it may only be done in rare occasions. It is suggested to regularly rouse the yeast in a forced ferment test to ensure a complete fermentation even with well flocculating yeasts.
- **fermentation time:** This is more practical for lager brewing than for ales. During the end of the primary fermentation and the long and cold lagering phase a very slow fermentation is in progress. By taking the beer off the yeast once the desired attenuation has been reached, a targeted attenuation/limit of attenuation difference can be achieved if the attenuation was already close enough when lagering started. The latter is important since only a few percent of attenuation gain can be expected during lagering. When considering removing a beer from the yeast it is important that green beer flavors like diacetyl and acetaldehyde have sufficiently been reduced since their reduction relies on the yeast.
- **mashing/wort composition:** Yes, mashing also has an affect on the fermentation and yeast performance. Lower ratios of maltotriose and higher content of glucose and maltose lead to better attenuation by the yeast since the yeast can metabolize the simple sugars more easily.

## practical means of affecting the attenuation

Although there are many factors that effect attenuation only a few are generally considered practical means for attenuation control by the the home brewer. Keep in mind that you want to keep the other attenuation affecting parameters the same to observe only the change from the parameter that you wanted to change.

- use the **saccharification rest length and temperature** to fine-tune the limit of attenuation for a given recipe and mash procedure. The other mash parameters should be optimized for conversion performance.
- Choose a **yeast strain** that suits the beer style and ensure adequate pitching rate as well as good yeast health. The flocculation and other fermentation characteristics of the yeast should get you the desired difference between attenuation and limit of attenuation.

# Appendix

## converting apparent to real attenuation

By using this formula for the real extract [Realbeer]:

$$RE = 0.1808 * OE + 0.8192 * AE$$

One can find the relationship between real attenuation (RA) and apparent attenuation (AA) as follows

$$RA = 1 - RE/OE$$

$$RA = 1 - (0.1808 * OE + 0.8192 * AE)/OE$$

$$RA = 1 - 0.1808 - 0.8192*AE/OE$$

$$RA = 0.8192 + 0.1808 - 0.1808 - 0.8192*AE/OE$$

Whith  $AA = 1 - AE/OE$  the real attenuation can be expressed as:

$$RA = 0.8192 * AA$$

## Sources

[Realbeer] realbeer.com Attenuation and related formulae  
(<http://www.realbeer.com/spencer/attenuation.html>)

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